Isochores Gibbs

A Bayesian Statistical Analysis on the GC content of chromosomes

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Course: Bayesian Inference

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Abstract:

This project implements a Bayesian changepoint model to analyze the GC content of a chromosome segment (chr3) using Gibbs sampling.

The dataset consists of CG base counts across 100 consecutive windows of 5,000 bases each. We assume a two-isochore model with an unknown changepoint t, where the probability of observing a CG base shift from p1 to p2. Conjugate Beta (0.5, 0.5) priors are placed on p1 and p2, and a discrete uniform prior is assigned to the changepoint t.

A Gibbs sampler is constructed to sample from the joint posterior distribution of (p1, p2, t). The algorithm successfully converges, and posterior samples reveal clear evidence of a structural shift in GC content, supporting the two-isochore hypothesis.

Visual diagnostics, including trace plots and histograms, confirm the stability and efficiency of the sampler. The results align with earlier Bayesian model selection conclusions, which decisively favored the two-isochore model over a single-isochore model for all chromosomes analyzed.

Keywords: Gibbs sampling, changepoint model, isochores, Bayesian inference, MCMC, GC content

Introduction

The analysis of GC content along chromosomal sequences is a fundamental problem in genomics, offering insights into genome organizations, evolutionary biology, and functional genomics. Isochores—long DNA segments with relatively homogeneous GC content—are key structural features in many eukaryotic genomes. Detecting shifts in GC proportions can reveal boundaries between distinct isochores, providing evidence of genomic heterogeneity.

In this project, we focus on the third dataset from a larger study of five chromosome segments, each comprising 100 consecutive windows of 5,000 bases. Previous Bayesian model comparison strongly favored a two-isochore model over a single-isochore model for all chromosomes, including chr3. To further investigate the nature and location of the changepoint, we adopt a Bayesian changepoint framework and implement a Markov Chain Monte Carlo (MCMC) method—specifically, a Gibbs sampler—to perform full posterior inference on the model parameters.

The model assumes a single changepoint t, partitioning the sequence into two regions with distinct CG probabilities p1 and p2. We use conjugate Beta priors (Jeffrey's prior) to facilitate analytical full conditional distributions, enabling an efficient Gibbs sampling scheme. This approach allows us to not only estimate the CG proportions in each region but also to infer the most probable location of the changepoint with associated uncertainty.

Data

The file *dataproject.R* contains five datasets, each corresponding to a segment of a chromosome. Each dataset consists of 100 consecutive windows, where each window contains 5000 DNA bases.

For every window, the recorded value is the number of bases of type C or G (CG count).

This allows to model each observation as:

$$X_i \sim \text{Binomial}(n = 5000,\ \theta_i),$$

Where

X_i is the number of CG bases in window i
$n = 5000$ is the number of bases per window,
θ_i is the underlying CG proportion for that region of the chromosome.

Thus, each chromosome dataset contains 100 binomial observations, summarizing local GC content.

Methods

Model Specification

We assume a two-isochore changepoint model for the CG counts in chromosome 3. Let xi denote the number of C or G bases in window i=1, ..., 100, each window containing n=5000 bases. The model is:

$$x_1, \ldots, x_t \sim \text{Binomial}(n, p_1),$$
$$x_{t+1}, \ldots, x_l \sim \text{Binomial}(n, p_2),$$

where:

- t is the unknown changepoint ($t \in \{1, \ldots, l-1\}$),
- p_1 and p_2 are the CG probabilities before and after the changepoint, respectively.

Priors:

To maintain objectivity in the absence of strong prior information, we used uninformative priors:

- $p_1, p_2 \sim \text{Beta}(0.5, 0.5)$ (Jeffreys prior),
- $t \sim \text{Discrete Uniform}(1, l-1)$.

Joint Posterior Calculation (up to a constant):

$$p(p_1, p_2, t \mid \mathbf{x}) \propto \text{Beta}(p_1 \mid a_1 + S_t, b_1 + nt - S_t) \times$$
$$\text{Beta}(p_2 \mid a_2 + (S - S_t), b_2 + n(l-t) - (S - S_t)) \times \mathbb{I}(t \in \{1, \ldots, l-1\})$$

Conditional Posteriors:

Conditional posteriors for the problem parameters p1, p2, t are derived from the joint posterior by considering only the terms with dependance on the parameter of interest in each case.

- $p_1 \mid \mathbf{x}, t \sim \text{Beta}(a_1 + S_t,\ b_1 + nt - S_t)$,
- $p_2 \mid \mathbf{x}, t \sim \text{Beta}(a_2 + (S - S_t),\ b_2 + n(l-t) - (S - S_t))$,

$$P(t|p_1, p_2, \mathbf{x}) = \frac{\left(\frac{p_1}{1-p_1}\right)^{S_t} \left(\frac{p_2}{1-p_2}\right)^{S'_t}}{\sum_{j=1}^{l-1} \left(\frac{p_1}{1-p_1}\right)^{S_j} \left(\frac{p_2}{1-p_2}\right)^{S'_j}}$$

where $S_j = \sum_{i=1}^{j} x_i$ and $S'_j = \sum_{i=j+1}^{l} x_i$.

When implementing the above in R, the conditional distribution for t is prone to overflow errors. To avoid such bottlenecks in computation, loglikelihood and “tricks” for numerical stability are being employed.

Gibbs Sampling Algorithm

We implemented a Gibbs sampler that iteratively draws from each full conditional:

1. Initialize p1, p2, t from their priors.
2. For each MCMC iteration:
   a. Sample p1, p2 from its Beta full conditional.
   b. Sample p1, p2 from its Beta full conditional.
   c. Sample t using the normalized posterior probability over all possible changepoints, computed on the log scale for numerical stability.
3. After a burn-in period, store samples for inference.

Implementation and Diagnostics

The sampler was implemented in R. Convergence was assessed via trace plots of the parameters. Posterior summaries, including histograms and estimated changepoint distributions, were computed from the post-burn-in samples.

Results

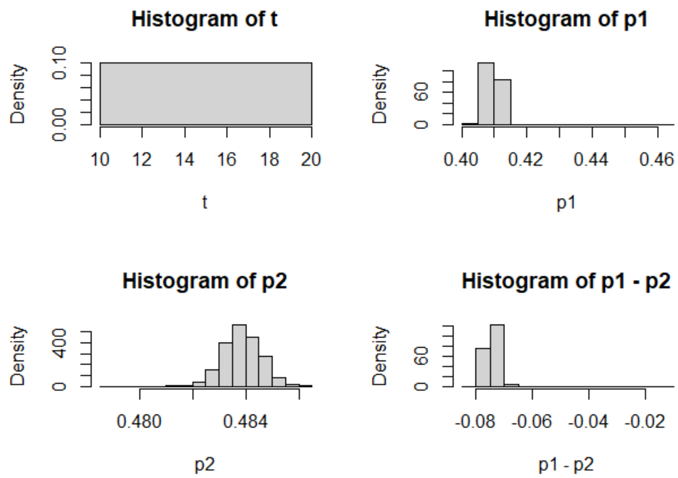
The Gibbs sampler was run for 5,000 iterations following a burn-in period. Convergence was assessed through trace plots of the parameters p1, p2, and t, which showed good mixing and stationarity (Figure 2). The posterior samples provide clear evidence of a structural changepoint in the GC content of chromosome 3.

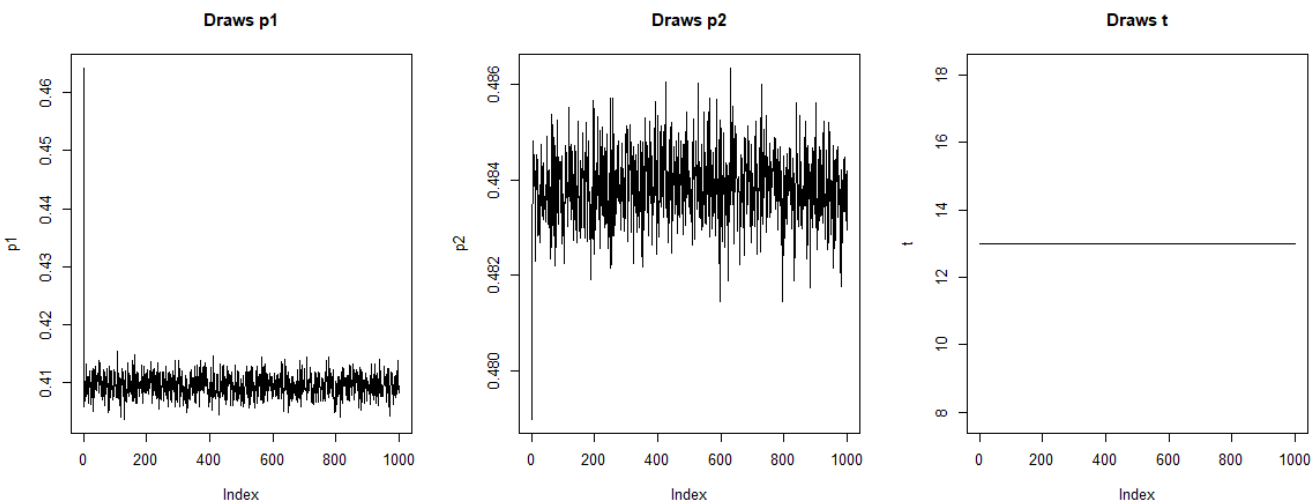
Changepoint Location

The posterior distribution of the changepoint t is strongly concentrated around window 1 (Figure 1), with the highest posterior probability at t=13. High concentration of the posterior distribution of t in 13 suggests a well-defined transition point in the chromosome, separating a lower-GC region (windows 1–13) from a higher-GC region (windows 14–100).

Model Validation

The posterior distribution of p1-p2 is entirely negative and excludes zero, providing strong Bayesian evidence against a single-isochore model. This aligns with earlier Bayes factor results, which decisively favored the two-isochore model.


Histogram of t
Density
0.00
0.10
10
12
14
16
18
20
t
Histogram of p1
Density
0
60
0.40
0.42
0.44
0.46
p1
Histogram of p2
Density
0
400
0.480
0.484
p2
Histogram of p1 - p2
Density
0
60
-0.08
-0.06
-0.04
-0.02
p1 - p2


Draws p1
p1
0.41
0.42
0.43
0.44
0.45
0.46
0
200
400
600
800
1000
Index
Draws p2
p2
0.480
0.482
0.484
0.486
0
200
400
600
800
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Index
Draws t
t
8
10
12
14
16
18
0
200
400
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Index